

MOMO SMS PROCESSING ERD

ERD LINK: [View ERD Diagram](#)

ERD Design Rationale & Justification: The MoMo SMS Data Processing System database architecture is designed to translate unstructured Mobile Money SMS transaction records into a 3NF-compliant relational database. The schema prioritizes data normalization, referential integrity, query performance, and analytical scalability.

Entity Identification & Data Normalization: The schema centers on four domain entities: **Users**, **Transactions**, **Transaction_Categories**, and **System_Logs**. Normalizing user details into a dedicated Users entity eliminates redundant sender and receiver data. Each user can participate in multiple transactions as a sender or receiver, represented through dual 1:M relationships (sender_id and receiver_id foreign keys referencing the Users primary key).

Transaction Categorization & Analytical Aggregations: Transactions are classified by linking each transaction record to a Transaction_Categories entity via a 1:M foreign key relationship (category_id). Pre-defining explicit categories (e.g., Peer Payment, Cash In, Airtime, Transfer) optimizes SQL aggregation queries, accelerates summary reporting, and ensures consistent transaction classification.

Resolving Many-to-Many (M:N) Relationships: To support individual user notification preferences across various transaction categories, a Many-to-Many (M:N) relationship exists between Users and Transaction_Categories. This is cleanly resolved using a dedicated junction table, User_Category_Preferences, featuring composite primary keys (user_id, category_id) and explicit foreign key constraints, preserving 3NF relational purity.

Auditability & System Health Monitoring: The System_Logs entity isolates operational logging, XML parsing exceptions, and dead-letter queue events from financial transaction storage. This architectural decoupling ensures high-volume diagnostic logging does not degrade transaction processing speeds while maintaining complete system auditability.

Overall, this entity-relationship design guarantees strict referential integrity, eliminates data anomalies, and provides an efficient foundation for enterprise mobile money analytics.

